Where does the infill housing go? Addition versus division

Thomas Davidoff and Tsur Somerville
Sauder, UBC

March 24, 2026

Overview

- Introduce a puzzle regarding upzoning for "infill housing"
  - New density allowances in "single family" zones *should* be most valuable in high-priced areas
  - But takeup of infill options leans towards lower-priced areas
- Resolve the puzzle with a model
  - "addition" (more square feet) vs "division" (more units)
  - Land quality $\Rightarrow$ wealth $\Rightarrow$ less gain to division
  - A lot of infill upzoning is division vs addition
- Data: upzonings in Vancouver, Minneapolis, ~~Portland~~
- Empirical contributions:
  - Model assumption is true of single family zones:
    - Price/sqft highly correlated with elasticity of price wrt sqft
  - Further evidence of negative sorting in division-only upzonings
  - Some evidence of more positive sorting in mixed addition/division upzonings

Single vs multifamily zoning a very old question
Brooks (1981) imagining Roman Senate

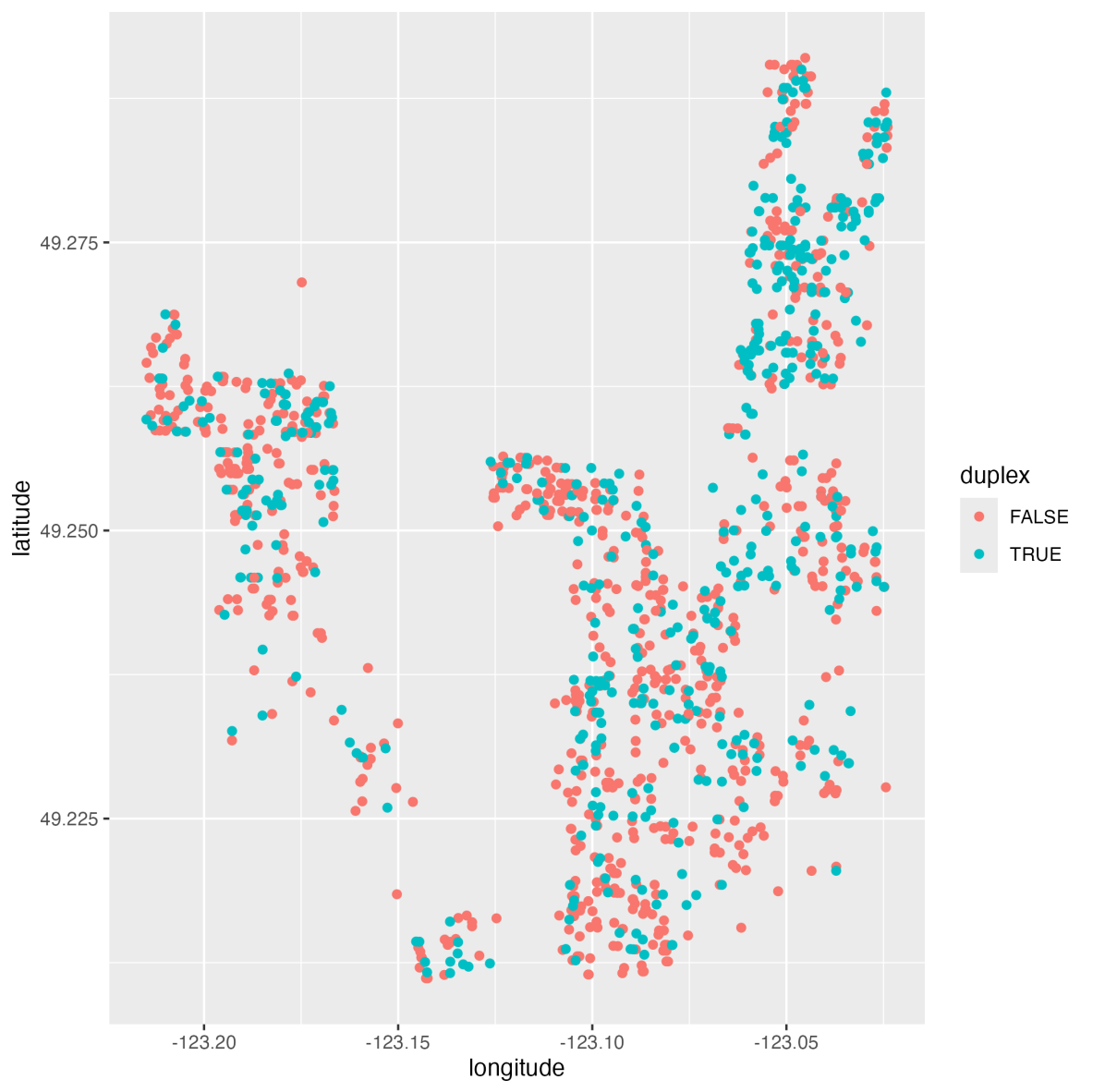
Leader of Senate: All fellow members of the Roman senate hear me. Shall we continue to build palace after palace for the rich? Or shall we aspire to a more noble purpose and build decent housing for the poor? How does the senate vote?
Entire Senate: F— THE POOR!

Zoning deadweight loss greatest in best locations

- Glaeser and Gyourko (2003) and Brueckner and Singh (2008): compare marginal price to cost /sqft
- Cities' natural shape: tallest, densest use closest to downtown
- For structure MC = price, need a very tall building downtown
- For land and structure bang-for-buck equality, need low density in suburbs
- Policy guidance re: upzoning to improve affordability:

The model in this report identifies the lots most ripe for redevelopment under the specified density scenarios, where the gains for converting from the exist- ing single family properties are highest, as the size of the wedge. Consistent with the basic model in urban economics, these are areas where land values for single family lots are highest. In Metro Vancouver, the City of Vancouver (especially its west side), the North Shore, Richmond and Burnaby stand out. These results correspond well with expectations that high amenity ar- eas closest in to the metropolitan centre would receive the greatest attention for higher density redevelopment if it were enabled.

Thus when Vancouver allowed 2 homes vs 1 per lot. . . Oh no!



Not just Vancouver duplex where infill goes to low θ

- Vancouver laneway (Davidoff, Pavlov, Somerville, 2022)
- ADUs in California (Marantz-Elmendorf and Kim, 2023; Brueckner and Thomaz, 2024)
- Duplexes and triplexes in Seattle (Huang et al, 2023)
- Portland's Residential Infill Project (RIP) (Dong and Hansz, 2019)
- Maybe we need a better model of infill location

A model of infill location

- "primitives"
  - $p(s, \theta)$: price per square foot of structure
    - Size $s = \frac{f}{n}$
    - f: floor area
    - n: number of units
    - Location θ
    - Lot size fixed at 1 unit of land
    - Assume: $p_\theta > 0$.
    - Reduced form sorting (single family!): $p_{s\theta} > 0$
  - Profit from raw land development $\pi = f\left[p(s, \theta) - c\right]$
    - c: cost per sqft of structure, constant for now
    - f not necessarily freely chosen

Positive sorting in theory?

- Standard urban model, for each type,

$$p_\theta \approx \frac{\text{Willingness to pay for } \theta}{\text{sqft}}$$

- Tradeoff: rich guys with more numerator also want denominator
- Can have poor people downtown apartments and rich people suburban detached homes
- But strict zoning deletes the denominator effect
- So only numerator $\Rightarrow$ expect $+$ sorting

Upzoning comparative statics addition versus division

- Addition: allow extra f
  - $\frac{\partial \pi}{\partial f} = p - c + sp_s$
  - $\frac{\partial^2 \pi}{\partial f \partial \theta} = p_\theta + sp_{s\theta} > 0$: more profitable in better locations
- Division: allow more n
  - $\frac{\partial \pi}{\partial n} = -s^2 p_s$
  - $\frac{\partial^2 \pi}{\partial n \partial \theta} = -s^2 p_{s\theta} < 0$: more profitable in worse locations

Upzonings in Vancouver

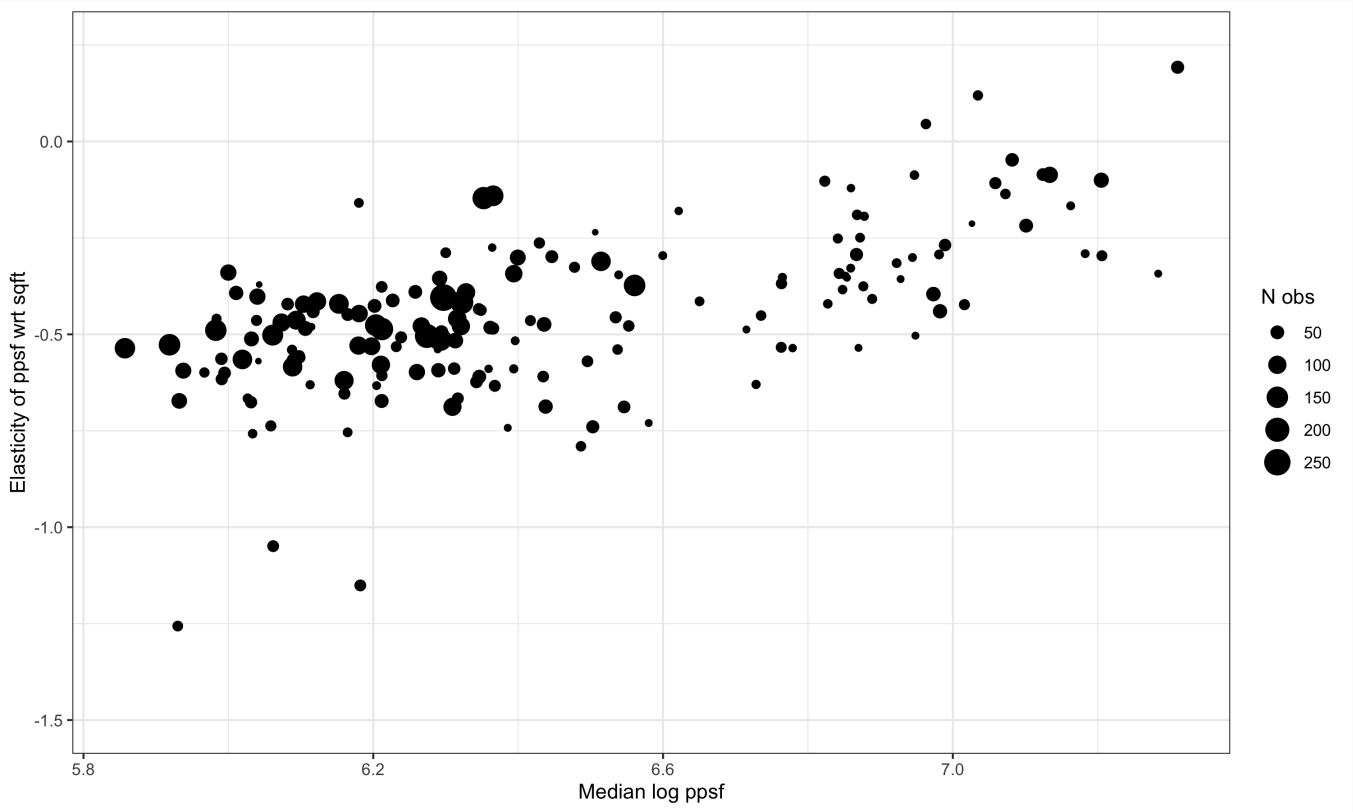
- Predominant Zoning: R1-1 now, used to be several R zones
- Baseline was single family detached homes
- 1980s-2004: Basement suites permitted citywide
- 2009: laneway homes permitted in single family homes
  - Add square feet, but must be rented, lose garage space
  - Laneway allowed .16-.25 FSR
- Sept 2018: duplexes allowable throughout R zones
  - No extra square feet (FSR .6-.7), but can sell 2 vs 1 home
  - Generally *can't* add a laneway
  - Hence division and *negative* addition
- 2023: multiplexes allowed
  - 4-6 units permitted depending on lot size
    - Much more attractive on wider lots
    - Roughly 50% of Vancouver lots 33' standard
  - FSR goes from .7 to 1.0
  - Hence more division, more addition than duplex
  - Also: single family down to .6 FSR, .7 for duplex

Data Sources

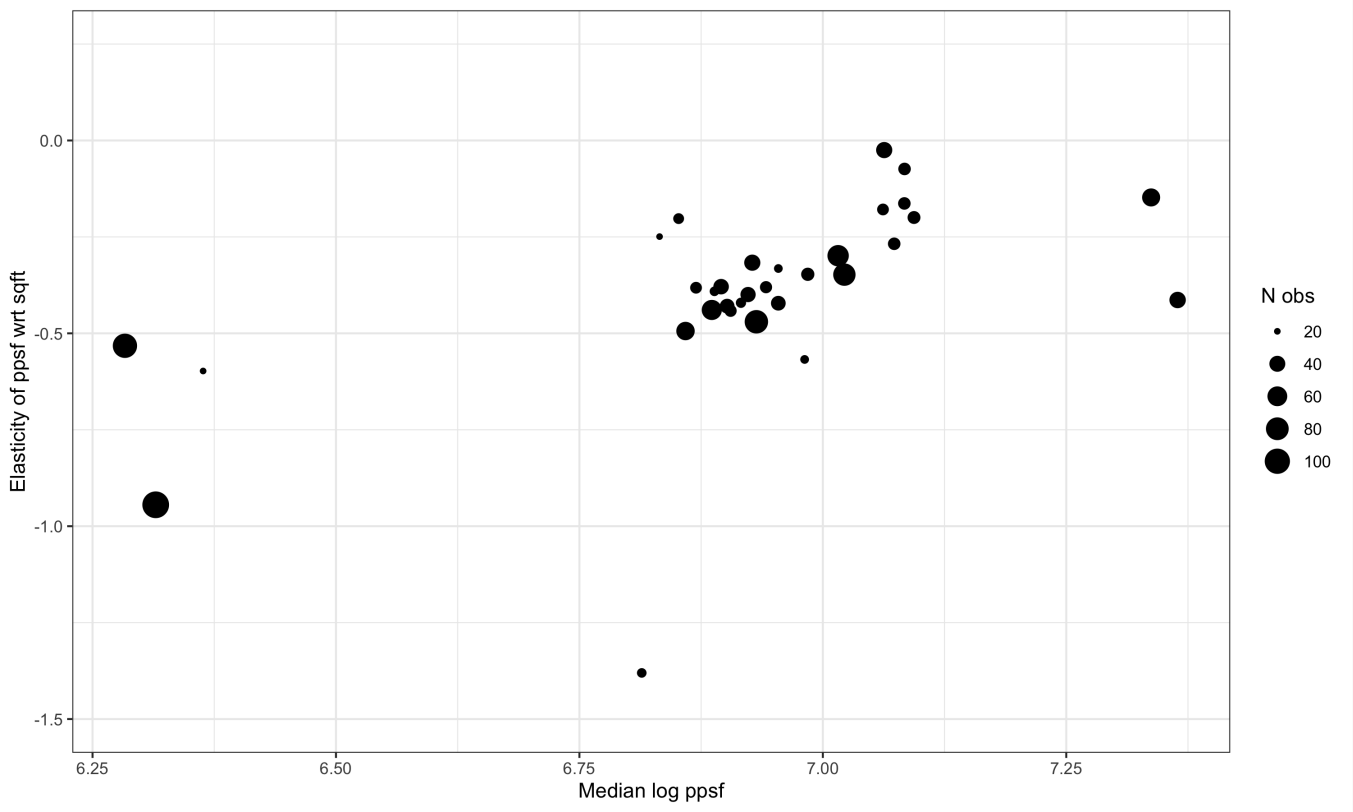
City	Subject	Data Source	Coverage/Use
Vancouver	Building Permits by	City of Vancouver	2019-2025 (lane
			2024-2025 (lane
Vancouver	Zoning	City of Vancouver	Current R1-1 zo
Vancouver	Transactions	B.C. Assessment	2017-2019 for b
Vancouver	Home and lot Characteristics	B.C. Assessment	All
Minneapolis	Building permit	City of Minneapolis	2020-2025 (post
Minneapolis	Zoning	City of Minneapolis	Interior 1,2, and
Minneapolis	Transactions	Attom	2016-2020
Minneapolis	Home and lot Characteristics	Attom	

Is it true that $p_{s\theta} > 0$?

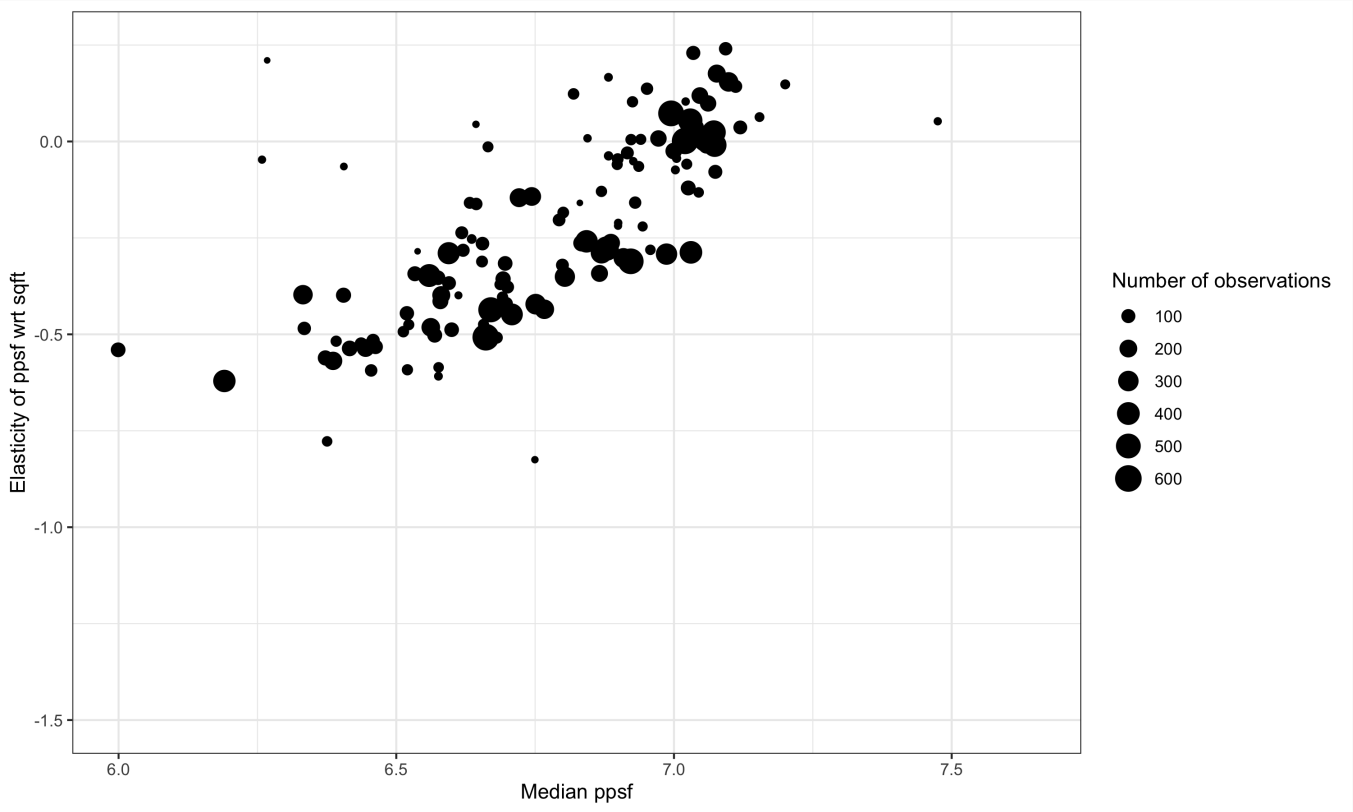
- BCA Single Family sales 2023-2024, Metro Vancouver
- Observation: a neighbourhood
- Elasticity < -1 means $p_s < 0$, > 0 means want lot assembly
  - Both implausible



Price per square foot and elasticity: duplexs



Price per square foot and elasticity: condos
Constant elasticity in size not a good fit
Supply matters, too (competition should make close to 0)



Spec vs custom: single vs not

- Spec(ulative): developer buys, builds, sells to market
- Custom: rich guy buys, hires builder, holds property
- Hard to do custom other than single family
- Exercise: match CoV new build permit to subsequent sale, 2017-2021:
  - Duplex 41 no, 77 yes
  - Single 1884 no,706 yes
- Custom likely fancier and not observed in sales data
- Elasticity hence likely biased down, more so West Side
- Note: can't do recent time slope of custom share: mechanical

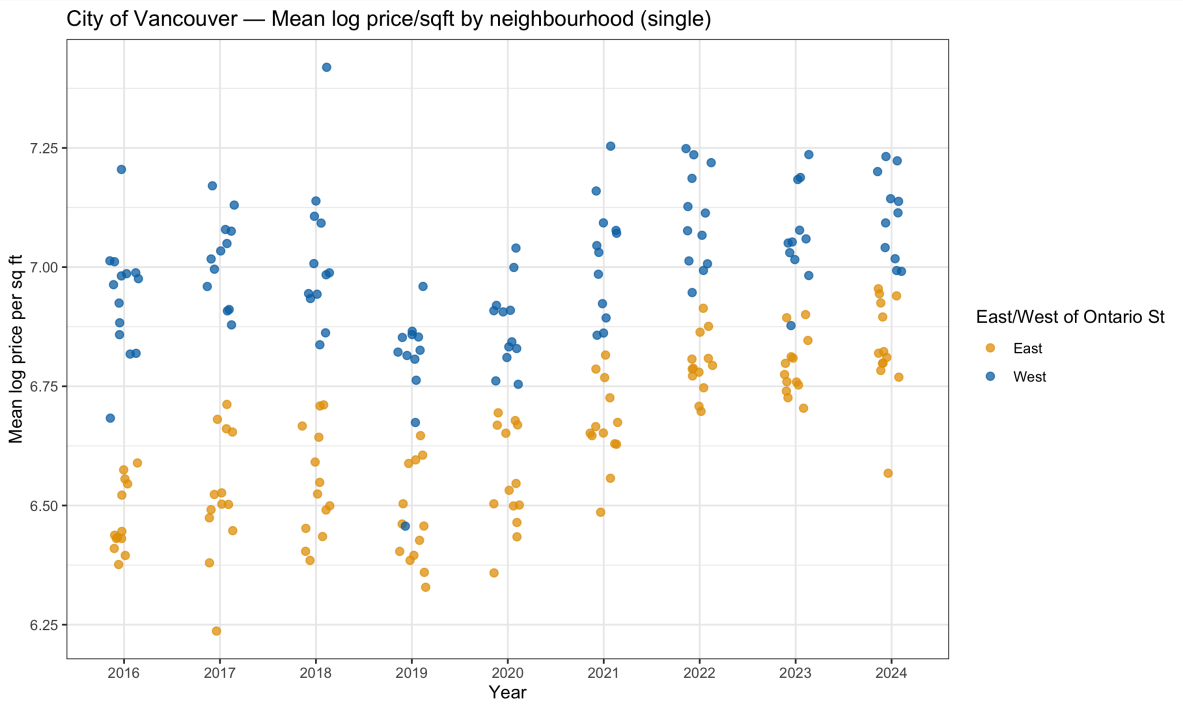
Not surprisingly, $p_{s\theta}^{condo} < p_{s/theta}^{duplex} < p_{s\theta}^{sf}$

- Condos built to maximize $p - c$
- Singles too big
- Duplexes presumably ease waste of space
- Elasticity < -1 means $p_s < 0$, > 0 means want lot assembly
    - Both implausible

City of Vancouver unit regressions by type

Dependent Variable:	logppsf		
Model:	(1)	(2)	(3)
Variables			
logsqft	-0.571***	-0.651***	-0.079***
	(0.002)	(0.012)	(0.002)
Fixed-effects			
year	Yes	Yes	Yes
NEIGHBOURHOOD	Yes	Yes	Yes
Fit statistics			
Observations	99,761	9,826	201,358
R^2	0.77528	0.58459	0.69394
Adjusted R^2	0.77480	0.57858	0.69374

Note dramatic price compression over time (foreign/spec?)
Dot is a neighbourhood x year



East/West Side and infill takeup over time: $\approx$ 33' lots

East/West Side and infill takeup over time: 50'+ lots

Neighbouhood x year Correlations: 20+, 33+ 40+ obs

	elasticity	intercept	duplex_premium	shareDuplex
elasticity	1.00	-0.99	-0.89	-0.11
intercept	-0.99	1.00	0.94	0.08
duplex_premium	-0.89	0.94	1.00	0.01
shareDuplex	-0.11	0.08	0.01	1.00

	elasticity	intercept	duplex_premium	shareDuplex
elasticity	1.00	-0.99	-0.84	-0.15
intercept	-0.99	1.00	0.91	0.16
duplex_premium	-0.84	0.91	1.00	0.14
shareDuplex	-0.15	0.16	0.14	1.00

	elasticity	intercept	duplex_premium	shareDuplex
elasticity	1.00	-0.99	-0.91	-0.22
intercept	-0.99	1.00	0.95	0.21
duplex_premium	-0.91	0.95	1.00	0.17
shareDuplex	-0.22	0.21	0.17	1.00

Duplex vs single prices by East/Lot Width/Pre Post Percentage vs dollar difference, 2Duplex vs 1 single

east	widthCat	prePost	use	V1
FALSE	wide	pre	single	3746968.54
FALSE	wide	pre	duplex	2330138.76
TRUE	wide	pre	single	1303430.99
TRUE	wide	pre	duplex	1347562.60
FALSE	wide	post	single	4766580.72
FALSE	wide	post	duplex	2786995.62
TRUE	wide	post	single	2018256.53
TRUE	wide	post	duplex	1884056.27
FALSE	standard	pre	single	2684545.86
FALSE	standard	pre	duplex	1752426.75
TRUE	standard	pre	single	1558812.59
TRUE	standard	pre	duplex	1264738.68
FALSE	standard	post	single	3357192.98
FALSE	standard	post	duplex	2159571.82
TRUE	standard	post	single	2125509.22
TRUE	standard	post	duplex	1562854.13

Minneapolis, MN

- 2018 Council vote, effective 2020
- 3 regions: Interior-1, Interior-2, Interior-3
- All newly permit duplex and triplex vs SF only
- Interior-1: FSR (unch) 0.5 , max height 2.5 storeys (28 ft)
- Interior-2: FSR (unch) 0.5 , max height 2.5 storeys (28 ft)
- Interior-3: FSR up: 0.6 for duplexes, 0.7 for triplexes, max height 3 storeys (42 ft)

Minneapolis Attom and permit data analysis
Post-reform period only

- For each zip code compute a mean price
- And an elasticity
  $\ln \frac{P_{izt}}{\text{sqft}_{izt}} = \beta_{0z} + \beta_{1z} \ln \text{sqft}_{izt} + \gamma X_{izt} + \delta_t + \epsilon_{izt}$
- These statistics computed 2 years pre-refom
- With permit data, compute fraction by zip code of permits
  - Duplex or triplex vs single family
  - In post-reform period only
- Study correlations
- Expect: Interior-1 and -2: division only - sorting
- Interior-3: addition and division - more positive sorting
- Only 31 zips total (11 in UN-3)

Sorting of infill in Minneapolis

All affected Zips

	elasticity	price_level	propensity	medianIncome
elasticity	1.00	0.59	0.28	0.30
price_level	0.59	1.00	0.03	0.69
propensity	0.28	0.03	1.00	-0.14
medianIncome	0.30	0.69	-0.14	1.00

Interior-3 zone

	elasticity	price_level	propensity	medianIncome
elasticity	1.00	0.61	0.32	0.31
price_level	0.61	1.00	0.24	0.82
propensity	0.32	0.24	1.00	0.01
medianIncome	0.31	0.82	0.01	1.00

Interior-1 and 2 zones

	elasticity	price_level	propensity	medianIncome
elasticity	1.00	0.58	0.36	0.29
price_level	0.58	1.00	-0.05	0.64
propensity	0.36	-0.05	1.00	-0.30
medianIncome	0.29	0.64	-0.30	1.00

Conclusions

- Infill zoning often more "division" than "addition"
- Fact + model + assumption $\Rightarrow$ negative sorting
- The assumption is true:
  - More expensive \$psqft more $\frac{\partial \log p}{\partial \log s}$
- Empirics: repeat usual negative sorting in:
  - Vancouver duplex in division-only era
  - Minneapolis division-only zones
- Some evidence with small samples of more + sorting with addition
  - Latter-day duplex if no laneway
  - Multiplex 2025
  - Interior-3 in Minneapolis
- Our wish: figure a way to get more power in analysis of small N areas